Multi-objective Cooperative Control of Vehicle Platoon Based on Fuzzy MPC

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***Abstract*—To simultaneously optimize multiple performances of the cooperative adaptive cruise control (CACC) system, a fuzzy MPC control strategy is proposed to achieve multi-objective cooperative control for electric vehicle platoons. The longitudinal kinematic analysis of the vehicle platoon is conducted to establish the platoon model. Based on the analysis of system performance, safety, comfort, and following performance are selected as the system's performance indexes, and a comprehensive performance function for the MPC is constructed. Fuzzy control theory is employed to optimize the weight parameters, enabling the CACC system to adapt to different conditions. The effectiveness and reliability of the fuzzy MPC control strategy are verified through co-simulation using Carsim and Simulink.**

***Index Terms*—fuzzy MPC control, vehicle platoon, intelligent connected vehicle, cooperative adaptive cruise control.**

I. Introduction

Vehicle platooning has great advantages in enhancing road safety, improving highway management efficiency, and optimizing fuel economy [1-2]. Therefore, in recent years, research on vehicle platoon systems has received increasing attention in the literature. Vehicle platoon control is based on V2X communication technology, and its main function is to enable multiple vehicles to travel in the same lane at a safe distance and same velocity [3-4].

Research on vehicle platoon control can be traced back to the PATH project in the 1980s, in which the researchers in this project preliminarily explored key issues related to platoon control [5]. Subsequently, Japan initiated a series of projects on vehicle platoon control, the most representative of which was the Energy ITS project, which focused on truck platooning and developed evaluation methods for energy consumption optimization [6]. In Europe, to achieve safer, more efficient, and energy-saving travel, vehicle platoon control technology has been researched, mainly including the SCANIA-platoon project and SARTRE project [7].

With the development of vehicle-related technologies, vehicle platoon control has enhanced safety and comfort by introducing environmental perception elements, such as radar and cameras, to obtain the driving states of neighboring vehicles [8]. In [9], a control algorithm for stop-and-go ACC is proposed to expand the application of ACC to low-speed and high-density urban roadways. In [10], a multi-lane ACC system is designed by considering multiple control objectives including safety, following, and economy. Additionally, some researchers have integrated the ACC system with other functional systems to further expand its application. D Kim et al. combined ACC with the collision avoidance system to investigate the ACC strategy capable of performing emergency braking and lane-change maneuvers [11].

The primary distinction between CACC and conventional ACC lies in the presence of inter-vehicle communication, and CACC enables control vehicles cooperatively to achieve overall optimization [12]. Current research on CACC mainly focuses on four aspects: safety, comfort, economy, and traffic efficiency. In [13], predictive feedback control is used to improve driving safety in the presence of communication delays. In [14], a distributed model predictive control algorithm has been proposed, which introduced a braking force allocation strategy for the following vehicles in the platoon, aiming to promote comfort. In [15], a distributed controller is proposed that combines feedback control with an economic model predictive control approach to optimize fuel consumption for the vehicle platoon. B. Liu et al. proposed a distributed CACC control algorithm for vehicles near intersections, improving traffic efficiency by reorganizing vehicle platoons around the intersection [16]. However, few studies have simultaneously optimized multiple performance indexes for CACC systems. Furthermore, existing CACC research predominantly focuses on fuel-powered vehicles, with limited attention given to new energy vehicles. Therefore, this paper investigates CACC strategies for electric vehicles, comprehensively considering safety, comfort, and following performance as optimization objectives.

The main contribution of this work is the application of the fuzzy MPC method to achieve cooperative control for the electric vehicle platoon. The platoon model is established through kinematic analysis. A comprehensive performance function and constraints are constructed to coordinate multiple control objectives, including safety, following, and comfort. Fuzzy rules are designed to optimize the weight parameters for safety and comfort.

This paper is organized as follows: Section II designs the framework of the CACC system and establishes the vehicle platoon model. Section III proposes a control strategy based on the fuzzy MPC algorithm to enhance the adaptability of the CACC system to different driving scenarios. Section IV analyzes the experimental results obtained by co-simulation with Carsim and Simulink. Section V presents the overall conclusions.

II. Problem Formulation

A. System Framework

The CACC system in this paper is divided into two parts: the decision layer and the execution layer. The system framework is shown in Fig. 1.

The decision layer calculates the desired acceleration of the vehicle using the fuzzy MPC algorithm, taking into account the states of other vehicles in the platoon. The execution layer converts the desired acceleration into corresponding driving or braking control commands, adjusting the vehicle's driving state to approach the desired state, thereby achieving longitudinal control of the vehicle. In driving mode, the driving force of the vehicle is provided by hub motors, so the desired acceleration needs to be converted into the desired driving torque.

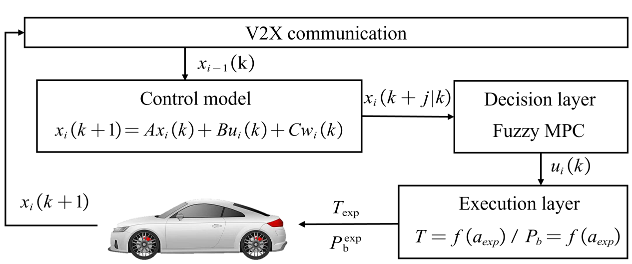


Fig. 1. CACC system framework.

$$T_{\mathrm{exp}}=\frac{\delta m a_{\mathrm{exp}}+Gf+\frac{1}{2}C_DA\rho v^2}{i_g i_0 \eta}r, \tag{1}$$

where T_{exp} is the desired driving torque, m is the mass of the vehicle, a_{exp} is the desired acceleration, δ is the mass conversion factor, G is the gravity force on the vehicle, f and C_D are the rolling and air resistance coefficients, A is the windward area of the vehicle, ρ is the air density, v is the velocity, i_g is the transmission ratio, i_0 is the differential ratio, η is the transmission efficiency, and r is the wheel radius.

Assuming that only mechanical braking is employed in brake mode, the braking force can be expressed as $F_b = -k_bP_b$. The brake mode requires converting the desired acceleration into the desired brake wheel cylinder pressure.

$$P_b=-\frac{\delta m a_{\mathrm{exp}}+Gf+\frac{1}{2}C_DA\rho v^2}{k_b}. \tag{2}$$

Since the driving mode and braking mode do not operate simultaneously during vehicle operation, frequent switching between the driving mode and braking mode affects comfort and causes some wear on the vehicle. Therefore, this paper designs a drive/brake mode switching region. This region is centered around the idle coasting acceleration a_{switch}, with an offset of $0.05\,m/s^2$. When the desired acceleration is above the upper bound of the region, the driving mode is activated, when it is below the lower bound of the region, the braking mode is applied.

B. Vehicle Platoon Control Model

Assume that a vehicle platoon consists of one lead vehicle and N following vehicles, where the lead vehicle is numbered 0, and the following vehicles are numbered from 1 to N. The state information of each vehicle in the platoon is transmitted to other vehicles via V2X communication. It is assumed that the communication is reliable, and communication delays are negligible.

This paper proposes a spacing strategy that integrates the overall motion trends of the platoon. The desired spacing of the ith vehicle is expressed as $d_i^{\mathrm{exp}} = d_c + t_hv_i$, where t_h is the headway, d_c is the correction term for the follow-up distance. The spacing error of the ith vehicle is defined as:

$$\varDelta d_i = d_i - d_i^{\mathrm{exp}}, \tag{3}$$

where $\varDelta d_i$ is the spacing error of the ith vehicle, d_i is the actual spacing between the ith vehicle and the preceding vehicle.

MPC solves a finite-horizon optimization problem at each sampling period, requiring the construction of a discretized model of the vehicle platoon. Based on longitudinal kinematic analysis and the first-order inertial hysteresis link of the actuator, the following linearized fifth-order node dynamic model can be established.

$$\begin{cases} d_i(k+1) = d_i(k) + v_i^{\mathrm{rel}}(k)T_s + \frac{1}{2}a_{i-1}(k)T_s^2 \\ \qquad -\frac{1}{2}a_i(k)T_s^2 \\ v_i(k+1) = v_i(k) + a_i(k)T_s \\ v_i^{\mathrm{rel}}(k+1) = v_i^{\mathrm{rel}}(k) + a_{i-1}(k)T_s - a_i(k)T_s \\ a_i(k+1) = \left(1 - \frac{T_s}{\tau}\right)a_i(k) + \frac{T_s}{\tau}u_i(k) \\ j_i(k+1) = -\frac{1}{\tau}a_i(k) + \frac{1}{\tau}u_i(k). \end{cases} \tag{4}$$

where T_s is the sampling period, τ is the time constant, v_{rel} is the relative velocity of the ith following vehicle to the preceding vehicle, j is the acceleration change rate. The discrete state-space equations can be established based on (4).

$$x_i(k+1) = Ax_i(k) + Bu_i(k) + Cw_i(k), \tag{5}$$

where $x_i(k) = \left[d_i(k), v_i(k), v_i^{rel}(k), a_i(k), j_i(k)\right]^T$ is the state vector, $w_i(k) = a_{i-1}(k)$ is the acceleration disturbance of the preceding vehicle, and A, B, C are coefficient matrices.

The main goal of vehicle platooning is to maintain the required safety distance between vehicles in the platoon and ensure that each vehicle follows the preceding vehicle at the same speed. Therefore, the vehicle platooning control should satisfy safety and followability. Additionally, comfort directly affects the satisfaction of the driver and passengers with the CACC system. Hence, safety, followability, and comfort are chosen as the performance indicators for the CACC system. These indicators can be mathematically expressed as follows:

$$\begin{array}{c} \varDelta d_i(k) \to 0, \quad v_i^{rel}(k) \to 0, \\ a_i(k) \to 0, \quad j_i(k) \to 0. \end{array} \tag{6}$$

Therefore, the comprehensive performance index $y_i(k)$ of the system is composed of the spacing error $\varDelta d_i(k)$, relative velocity $v_i^{rel}(k)$, acceleration $a_i(k)$, and its rate of change $j_i(k)$. The comprehensive performance index can be expressed as the following equation:

$$\begin{aligned} y_i(k) &= Dx_i(k) + E \\ &= \left[\varDelta d_i(k), v_i^{rel}(k), a_i(k), j_i(k)\right]^T \end{aligned} \tag{7}$$

where D and E are the coefficient matrices.

III. Longitudinal Vehicle Platoon Control

A. Constraints for Platoon Controller

Before designing the MPC controller, it is necessary to specify the constraints for each state variable and input variable.

a) Safety Spacing: A safe distance between the vehicle and in front of it should be maintained to reduce the risk of collision.

$$d_{\min} \leqslant d_i(k) \leqslant d_{\max}, \tag{8}$$

b) Longitudinal Speed: To ensure that the vehicle is traveling within the specified speed range, the vehicle speed must not exceed the maximum permissible speed.

$$v_{\min} \leqslant v_i(k) \leqslant v_{\max}, \tag{9}$$

c) Acceleration Range: The absolute values of the acceleration and its rate of change should not exceed a maximum allowable value to ensure both vehicle acceleration performance and passenger comfort.

$$a_{\min} \leqslant a_i(k) \leqslant a_{\max}, \tag{10}$$

$$j_{\min} \leqslant j_i(k) \leqslant j_{\max}, \tag{11}$$

$$u_{\min} \leqslant u_i(k) \leqslant u_{\max}. \tag{12}$$

In the MPC optimization process, hard constraints can often lead to infeasible solutions. To address this, penalty coefficients can be introduced to soften the above constraints. But the constraints on the safety spacing will not be softened due to safety considerations.

$$\begin{aligned} d_{min} + \lambda_1 d_{min\_s} &\leqslant d_i(k) \leqslant d_{\max} + \lambda_1 d_{\max\_s}, \\ v_{min} + \lambda_2 v_{min\_s} &\leqslant v_i(k) \leqslant v_{\max} + \lambda_2 v_{\max\_s}, \\ a_{min} + \lambda_3 a_{min\_s} &\leqslant a_i(k) \leqslant a_{\max} + \lambda_3 a_{\max\_s}, \\ j_{min} + \lambda_4 j_{min\_s} &\leqslant j_i(k) \leqslant j_{\max} + \lambda_4 j_{\max\_s}, \\ u_{min} + \lambda_5 u_{min\_s} &\leqslant u_i(k) \leqslant u_{\max} + \lambda_5 u_{\max\_s}. \end{aligned} \tag{13}$$

where λ_1, λ_2, λ_3, λ_4, λ_5 is the penalty factor for each constraint. To ensure the safety of driving, take λ_1 as 0.

B. Platoon Controller Based on MPC

MPC is used to predict the future dynamics of the system based on historical information and current inputs, and then compare these predictions with the desired trajectory. Noting N_p as the prediction horizon, and N_c represents the control horizon. Using the system's discrete state equation (5) and the comprehensive performance index equation (7), the state variable sequence and performance index sequence for the future time horizon N_p can be obtained.

$$X_i(k) = \widetilde{A}x_i(k) + \widetilde{B}U_i(k) + \widetilde{C}W_i(k), \tag{14}$$

$$Y_i(k) = \widetilde{D}x_i(k) + \widetilde{E}U_i(k) + \widetilde{F}W_i(k) + \widetilde{\mathrm{G}}. \tag{15}$$

where $U_i(k)$ is the control sequence, $W_i(k)$ is the acceleration disturbance sequence of the front vehicle, $X_i(k)$ is the state variable prediction matrix, $Y_i(k)$ is the performance index prediction matrix as follows:

$$X_i(k) = \begin{bmatrix} x_i(k+1 \mid k) \\ x_i(k+2 \mid k) \\ \vdots \\ x_i(k+N_p \mid k) \end{bmatrix} \quad Y_i(k) = \begin{bmatrix} y_i(k+1 \mid k) \\ y_i(k+2 \mid k) \\ \vdots \\ y_i(k+N_p \mid k) \end{bmatrix}$$

$$U_i(k) = \begin{bmatrix} u_i(k) \\ u_i(k+1) \\ \vdots \\ u_i(k+N_c-1) \end{bmatrix} \qquad W_i(k) = \begin{bmatrix} w_i(k) \\ w_i(k) \\ \vdots \\ w_i(k) \end{bmatrix}$$

where $x_i(k+j \mid k)$ and $y_i(k+j \mid k)$ represent the prediction of the state variables and the performance index at time step $k+j$, made by the MPC at time step k.

To comprehensively coordinate the control objectives of safety, following and comfort of the CACC system, the following comprehensive performance function is established to obtain the optimal control sequence.

$$
\begin{aligned}
J = & \sum_{j=1}^{Np} \left\| y_i\left(k+j|k\right) - y_i^{ref}\left(k+j|k\right) \right\|_{Q_i}^2 \\
& + \sum_{j=1}^{N_c} \left\| u_i\left(k+j-1\right) \right\|_{R_i}^2 + \sum_{j=1}^{5} \left\| \lambda_j \right\|_{\omega}^2
\end{aligned} \tag{16}
$$

$$
\begin{aligned}
s.t. X_{min} + X_{min\_s} \leqslant \mathit{\Phi} X_i\left(k\right) \leqslant X_{max} + X_{max\_s} \\
U_{min} + U_{min\_s} \leqslant U_i\left(k\right) \leqslant U_{max} + U_{max\_s}.
\end{aligned}
$$

where Q_i, R_i and ω are the weight matrices, $y_i^{ref}\left(k+j|k\right)$ is the performance index reference curve. From the formulation given in (16), the comprehensive performance function consists of three components. The first term ensures that each performance index converges to its desired reference value. The second term is for avoiding emergency braking and sudden acceleration. The third term imposes constraints on the penalty coefficients. Solving the above optimization problem at sampling moment k obtains the optimal control sequence, and the first control input in the sequence is selected as the desired acceleration and transmitted to the execution layer. The execution layer converts this desired acceleration into the corresponding driving or braking commands to adjust the driving state of the vehicle. At each sampling moment, the integrated performance function (16) is solved again for rolling optimization.

C. Weight Optimization Based on Fuzzy Control

The fixed weights of the MPC controller are insufficient for the CACC system to adapt to complex environments. Therefore, fuzzy control theory is employed to optimize the weight parameters of the MPC controller. It is necessary to determine whether the vehicle is in a safe following state, and based on different states, the safety weight ω_{safe} and comfort weight ω_{comf} are adjusted without changing the following weight ω_{vrel}.

This paper uses the headway and the reciprocal of the collision time as criteria for determining the safety status between the two vehicles. The headway is one of the important indexes for evaluating driving safety. By inverting the spacing strategy, the input T_H for the fuzzy controller can be obtained. The reciprocal of the collision time reflects the collision risk between vehicles. As indicated by (18), the larger the value of C_T, the lower the current risk of collision.

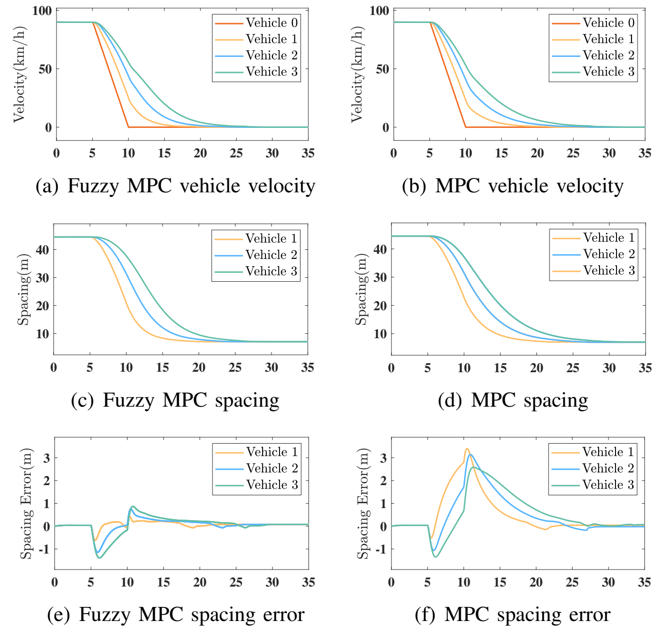
$$
T_H = \frac{d - d_0 + c v_{rel\_h}}{v}, \tag{17}
$$

$$
C_T = \frac{v_{rel}}{d}, \tag{18}
$$

After multiple adjustments, an exponential correction factor is chosen to modify the safety and comfort weights. The adjusted weight parameters are given in (19) and (20).

$$
\omega_{r_{safe}} = 2^{r_{safe}} \omega_{safe}, \tag{19}
$$

$$
\omega_{r_{comfort}} = 2^{r_{comf}} \omega_{comf}. \tag{20}
$$



(a) Fuzzy MPC vehicle velocity (b) MPC vehicle velocity

(c) Fuzzy MPC spacing (d) MPC spacing

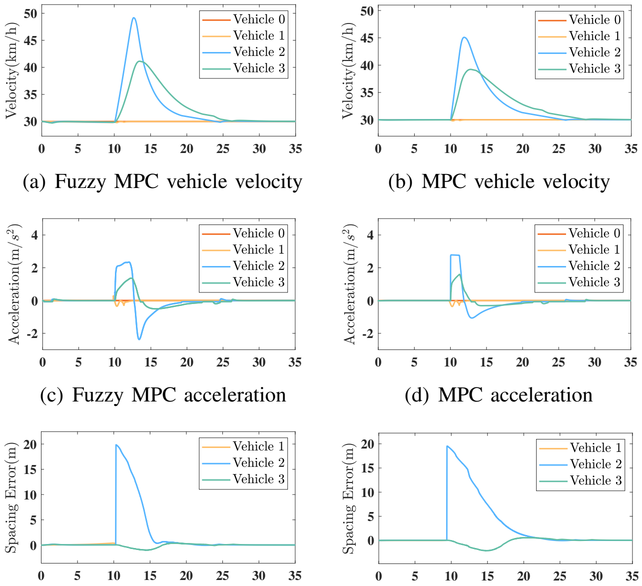
(e) Fuzzy MPC spacing error (f) MPC spacing error

Fig. 2. Emergency Braking Condition

A two-input, two-output fuzzy controller can be designed, with the inputs being T_H and C_T, and the outputs being r_{safe} and r_{comf}. The basic domains must first be determined when fuzzifying the input and output variables. Based on the analysis of driver behavior [17], the range for the input variable T_H is determined to be [0, 3], and the range for the reciprocal of the collision time C_T is [-0.3, 0.3]. After several rounds of tuning, the final range for both r_{safe} and r_{comf} is set to [-5, 5]. Once the basic domains for the input and output variables are determined, their ranges are fuzzified into the corresponding fuzzy sets $\{NB, NS, ZO, PS, PB\}$. Specifically, NB, NS, ZO, PS, and PB represent the degrees of deviation from zero in the negative or positive directions.

In addition, the membership functions for each variable need to be defined. Gaussian membership functions are used for the input variables T_H and C_T, while triangular membership functions are employed for the output variables r_{safe} and r_{comf}. According to fuzzy control theory, the core of a fuzzy controller is to establish the relationship between input and output variables through fuzzy reasoning, that is, by creating fuzzy rules. For the weight parameters of the MPC controller in this study, the fuzzy rules should follow these principles:

- When T_H is NB: The following state is considered dangerous, so the safety correction factor r_{safe} should be increased, while the comfort correction factor r_{comf} should be decreased.
- When C_T is NB: The collision risk between the vehicle and the preceding vehicle is high, so the safety correction factor r_{safe} should be increased significantly, while the comfort correction factor r_{comf} should be reduced.
- When T_H and C_T are ZO: The following state is considered safe, and therefore no adjustment is required



(a) Fuzzy MPC vehicle velocity
(b) MPC vehicle velocity
(c) Fuzzy MPC acceleration
(d) MPC acceleration
(e) Fuzzy MPC spacing error
(f) MPC spacing error

Fig. 3. Vehicle Leaving the Platoon Condition

for the safety and comfort weight parameters.

- When T_H is PB: The following state is very safe, so the safety correction factor r_{safe} should be appropriately reduced, while the comfort correction factor r_{comf} should be increased.
- When C_T is PB: The collision risk between the vehicle and the preceding vehicle is very low, so the comfort correction factor r_{comf} should be increased, while the safety correction factor r_{safe} should be decreased.

Based on the above principles, fuzzy rules for the safety modification coefficients and the comfort modification coefficients are established. These rules enable the inference of relationships between fuzzy input variables and fuzzy output variables. By applying the centroid method to defuzzify the fuzzy output variables, the relationships between the output variables and input variables can be determined.

IV. Simulation Validations

Co-simulation experiments are conducted using the Carsim and the Simulink module in MATLAB to verify the superiority of the proposed control strategy over the traditional MPC under different operating conditions.

This paper evaluates the control performance of the fuzzy MPC strategy compared to the MPC strategy in three common driving scenarios: emergency braking, vehicle leaving the platoon, and vehicle joining the platoon.

a) Emergency Braking Condition: The four-vehicle platoon travels at a constant high initial speed of $90\,km/h$. At $5\,s$, the lead vehicle begins decelerating at the maximum deceleration of $-5\,m/s^2$ until the vehicle stops. During the emergency braking process of the lead vehicle, the fuzzy MPC strategy shows a significant improvement in the safety correction factor, the spacing error is controlled within $1.5\,m$. In contrast, the MPC strategy allows the spacing error to reach up to $3.4\,m$ during emergency braking. Moreover, under the fuzzy MPC strategy, the spacing error converges to within $0.5\,m$ in $8\,s$, whereas the MPC strategy error decreases to $1.11\,m$ within the same time. This indicates that the fuzzy MPC control strategy provides a faster response speed under emergency braking conditions, and therefore has higher safety.

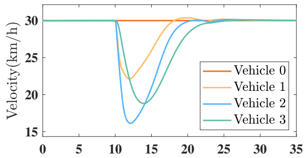
b) Vehicle Leaving the Platoon Condition: The lead vehicle cruises at a speed of $30\,km/h$. At $10\,s$, vehicle 1 leaves the four-vehicle platoon, and vehicles 2 and 3 accelerate to form a new three-vehicle platoon with the lead vehicle. Under this scenario, the fuzzy MPC control strategy significantly increases the comfort weight after vehicle 1 leaves the platoon due to the increased headway. It reduces the spacing error to within $1\,m$ in $5.4\,s$,compared to $6.63\,m$ for the MPC strategy in the same time. Additionally, the fuzzy MPC control strategy provides smoother acceleration compared to the MPC strategy in the acceleration process. This demonstrates that the fuzzy MPC strategy can quickly respond to the departure of a vehicle from the platoon and appropriately enhance comfort.

c) Vehicle Joining the Platoon Condition: The cruising speed of the lead vehicle is $30\,km/h$. At $10\,s$, vehicle 1 starts to drive into the platoon. Vehicles 2 and 3 quickly adjust their positions to form a new four-vehicle platoon. Under this condition, the fuzzy MPC strategy increases the safety correction factor after vehicle 1 joins the platoon, effectively reducing collision risk. In contrast, the MPC control strategy shows larger fluctuations in spacing error, which may increase the risk of collisions. Furthermore, during the process of reducing spacing error, the acceleration under the fuzzy MPC strategy is smoother compared to that of the MPC strategy. This indicates that the fuzzy MPC strategy outperforms the MPC strategy in terms of comfort and safety under the vehicle joining the platoon condition.

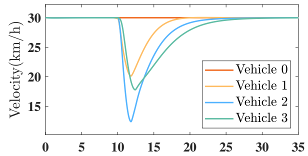
Analysis of the simulation results for the above three working conditions indicates that the CACC system can effectively regulate the driving speed and following distance of the vehicle platoon under different conditions. The fuzzy MPC strategy improves the adaptability of the CACC system to different driving environments compared to the MPC strategy, ensuring both safety and comfort.

V. Conclusions

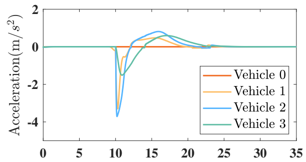
A CACC system for electric vehicles based on the fuzzy MPC algorithm is designed. Under the hierarchical CACC system framework, the vehicle node dynamics model is established. Safety, comfort, and following performance are selected as the performance indexes of the CACC system, and a comprehensive performance function is constructed. Fuzzy control theory is employed to optimize the safety and comfort weight parameters of the MPC controller, enabling the CACC system to adapt to different driving conditions. The effectiveness of the CACC system is verified under three typical driving conditions. Experimental results demonstrate the system's strong adaptability in these scenarios, confirming the effectiveness and feasibility of the fuzzy MPC algorithm.



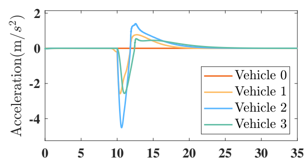
(a) Fuzzy MPC vehicle velocity



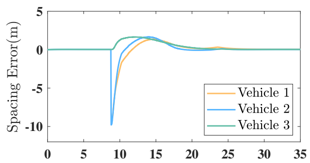
(b) MPC vehicle velocity



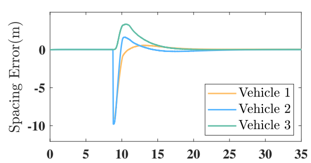
(c) Fuzzy MPC acceleration



(d) MPC acceleration



(e) Fuzzy MPC spacing error



(f) MPC spacing error

Fig. 4. Vehicle Joining the Platoon Condition

The next step of the study will be to investigate the effects of external disturbance and wireless communication delays on the control strategy, and to evaluate the control strategy through real-vehicle experiments.

References

[1] Y. Yin, X. Huang, S. Zhan, H. Gou, X. Zhang, and F. Wang, "Hierarchical energy management control based on different communication topologies for hybrid electric vehicle platoon," Journal of Cleaner Production, vol. 412, p. 137414, Aug. 2023.

[2] Y. Liu, C. Zong, C. Dai, H. Zheng, and D. Zhang, "Behavioral Decision-Making Approach for Vehicle Platoon Control: Two Noncooperative Game Models," IEEE Transactions on Transportation Electrification, vol. 9, no. 3, pp. 4626–4638, Sep. 2023.

[3] Y. Zhu et al., "Hierarchical Control of Connected Vehicle Platoon by Simultaneously Considering the Vehicle Kinematics and Dynamics," IEEE Transactions on Intelligent Vehicles, vol. 9, no. 1, pp. 1333–1345, Jan. 2024.

[4] Z. Qiang, L. Dai, B. Chen, K. Li, and Y. Xia, "Distributed model predictive control for heterogeneous vehicle platoon with unknown input of leading vehicle," Transportation Research Part C: Emerging Technologies, vol. 155, p. 104312, Oct. 2023.

[5] R. Rajamani, H.-S. Tan, B. K. Law, and W.-B. Zhang, "Demonstration of integrated longitudinal and lateral control for the operation of automated vehicles in platoons," IEEE Transactions on Control Systems Technology, vol. 8, no. 4, pp. 695–708, Jul. 2000.

[6] S. Tsugawa, "An Overview on an Automated Truck Platoon within the Energy ITS Project," IFAC Proceedings Volumes, vol. 46, no. 21, pp. 41–46, Jan. 2013.

[7] G. Toulminet, J. Boussuge, and C. Laurgeau, "Comparative synthesis of the 3 main European projects dealing with Cooperative Systems (CVIS, SAFESPOT and COOPERS) and description of COOPERS Demonstration Site 4," in 2008 11th International IEEE Conference on Intelligent Transportation Systems, Oct. 2008, pp. 809–814.

[8] R. Rajamani and C. Zhu, "Semi-autonomous adaptive cruise control systems," IEEE Transactions on Vehicular Technology, vol. 51, no. 5, pp. 1186–1192, Sep. 2002.

[9] K. Yi, J. Hong, and Y. D. Kwon, "A vehicle control algorithm for stop-and-go cruise control," Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering, vol. 215, no. 10, pp. 1099–1115, Oct. 2001.

[10] S. Tajeddin, S. Ekhtiari, M. Faieghi, and N. L. Azad, "Ecological Adaptive Cruise Control With Optimal Lane Selection in Connected Vehicle Environments," IEEE Transactions on Intelligent Transportation Systems, vol. 21, no. 11, pp. 4538–4549, Nov. 2020.

[11] D. Kim, S. Moon, J. Park, H. J. Kim and K. Yi, "Design of an Adaptive Cruise Control / Collision Avoidance with lane change support for vehicle autonomous driving," 2009 ICCAS-SICE, Fukuoka, Japan, 2009, pp. 2938-2943.

[12] T. Yang, C. Murguia, D. Nešić, and C. Lv, "A Robust CACC Scheme Against Cyberattacks via Multiple Vehicle-to-Vehicle Networks," IEEE Transactions on Vehicular Technology, vol. 72, no. 9, pp. 11184–11195, Sep. 2023.

[13] N. Bekiaris-Liberis, "Robust String Stability and Safety of CTH Predictor-Feedback CACC," IEEE Transactions on Intelligent Transportation Systems, vol. 24, no. 8, pp. 8209–8221, Aug. 2023.

[14] D. Pi, P. Xue, B. Xie, H. Wang, X. Tang, and X. Hu, "A Platoon Control Method Based on DMPC for Connected Energy-Saving Electric Vehicles," IEEE Transactions on Transportation Electrification, vol. 8, no. 3, pp. 3219–3235, Sep. 2022.

[15] M. Hu, C. Li, Y. Bian, H. Zhang, Z. Qin, and B. Xu, "Fuel Economy-Oriented Vehicle Platoon Control Using Economic Model Predictive Control," IEEE Transactions on Intelligent Transportation Systems, vol. 23, no. 11, pp. 20836–20849, Nov. 2022.

[16] B. Liu and A. El Kamel, "V2X-Based Decentralized Cooperative Adaptive Cruise Control in the Vicinity of Intersections," IEEE Transactions on Intelligent Transportation Systems, vol. 17, no. 3, pp. 644–658, Mar. 2016.

[17] S. Moon and K. Yi, "Human driving data-based design of a vehicle adaptive cruise control algorithm," Vehicle System Dynamics, vol. 46, no. 8, pp. 661–690, Aug. 2008.